Maria Granados

Senior AI/ML Engineer | Agentic Platforms & Enterprise AI

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Professional Summary

Senior AI Engineer with 7+ years turning ambiguous business problems into production ML and GenAI systems. Architects and delivers the layer other teams build on: enterprise agent platforms and harnesses, reusable AI skills, MCP integrations, self-service deployment and responsible-AI safeguards across the organisation's AI and data ecosystem. Designed how AI adoption and value are measured across 20+ platform products and a 16,000+ workforce, aligned to enterprise AI strategy and giving leadership the evidence base for continued investment. Outcomes include $1M+ vendor-tool replacement, $70K+ value in the first month and workflows reduced from weeks to seconds.

Core Skills

- **Agentic & GenAI platforms:** Enterprise agent platforms and harnesses, multi-agent orchestration, MCP servers/tools/Apps, reusable AI skills, context engineering, RAG, semantic/vector search, Bedrock, SageMaker, OpenAI SDK
- **AI platform engineering & LLMOps:** Self-service deployment, developer experience, Humanitec, Port.io, Docker, Kubernetes, CI/CD, MLflow, model versioning, evals, monitoring, A/B testing, design of experiments, AWS,Git, GitHub Copilot, Claude Code, OpenAI Codex, Java, R, Power BI, Streamlit
- **AI governance & responsible AI:** AI safeguards and guardrails, cybersecurity assurance, access controls, audit logging, vulnerability management, data protection, risk assessment and architecture trade-offs
- **Machine learning & data engineering:** Python, SQL, Snowflake, Databricks, PySpark, TensorFlow, PyTorch, transformers, scikit-learn, NLP, churn modelling, design of experiments, hyperparameter tuning, metadata-driven pipelines
- **AI strategy & enterprise enablement:** AI platform architecture, enterprise AI adoption and value measurement, reusable capability design, self-service enablement, data governance, semantic layers, frontline-to-C-suite influence and mentoring

Professional Experience

Senior AI Engineer (AI Platform)

Fortescue Metals Group | September 2025 – Present | Perth, Australia

- Architected an enterprise agent platform that turns a single natural-language request into governed, production-ready software: agents clarify requirements, generate plans, document assumptions and produce acceptance criteria before execution, then orchestrate delivery across enterprise systems including Atlassian, Snowflake and SharePoint via MCP.
- Built the Agent Skills Hub, a governed library of reusable AI capabilities within the enterprise agent harness, spanning software engineering, data engineering, document generation, deployment and assurance, reducing duplicated build effort across teams. One deployment skill compresses a workflow that could take up to a week with DevOps support into minutes, taking an application from localhost to enterprise SSO-enabled Humanitec deployment.
- Built an enterprise discovery catalogue that replaced $1M+ of incumbent tooling and was nominated by Snowflake for a global Data Driver award; delivered $70K+ in measurable value in the first month, reaching 44% adoption and 40% stickiness as a two-person team shipping fortnightly releases.
- Architected hybrid semantic search and RAG (OpenAI SDK, Snowflake Cortex, Databricks Mosaic, vector databases), cutting discovery time from weeks to seconds and automating workflows from 2 hours to 3 seconds at org-wide scale.
- Designed the Enterprise AI Adoption & Value framework defining how reach, adoption and business value are measured across every current and future platform product for the full 16,000+ workforce, aligned to the AI Director's strategic pillars and giving leadership the evidence base for continued AI investment.
- Embedded responsible-AI and cybersecurity safeguards into the agent harness, including access controls, audit logging, vulnerability management, data-protection controls and reusable assurance skills, with risks and architecture trade-offs documented for critical decisions.
- Built containerised AI pipelines (Docker, Kubernetes, CI/CD) with Humanitec and Port.io for developer self-service, and implemented LLMOps practices (model versioning, A/B testing, monitoring, automated retraining) with MLflow.

Senior ML Engineer / Data Scientist
P&N Bank | July 2023 – September 2025 | Perth, Australia

- Technical lead for all ML and GenAI initiatives bank-wide, mentoring the analytics team and uplifting data science capability for faster, more reliable delivery.
- Deployed the bank's first internal churn models across home loans and term deposits, surfacing a 30% churn-reduction opportunity that influenced product pricing strategy.
- Improved F1-score and PR-AUC by 20% on imbalanced datasets through factorial design of experiments and targeted hyperparameter tuning.
- Co-developed an NLP abuse-detection prototype using Hugging Face transformers on transaction data to protect vulnerable customers from financial abuse.
- Delivered Streamlit and Power BI automation saving $110K+ in FTE capacity, and built cloud-native Snowflake MLOps pipelines enabling versioning, CI/CD, and production monitoring.
- Drove organisation-wide data-literacy uplift saving an estimated $250K–$400K annually, partnering with C-suite, product, and frontline teams.

Data & AI Consultant
Deloitte | June 2022 – June 2023 | Perth, Australia

- Rescued a $1M high-risk data-migration project as technical lead, preventing $250K+ in further overruns and earning partner recognition for turnaround leadership.
- Led a cross-functional team of 10 to modernise a 20-year legacy database, enabling scalable analytics across departments.
- Designed metadata-driven ETL pipelines in Talend (Java, Regex) with CI/CD for maintainable ingestion of millions of records.
- Mentored team members in DevOps, Talend, Git, and pipeline debugging, lifting overall delivery capability.

Data Analyst
Curtin University | March 2021 – June 2022 | Perth, Australia

- Automated accreditation reporting for a Top 100 business school, cutting reporting time 90% (500+ hours saved biennially) and improving accuracy and compliance.
- Led end-to-end data migration and analytics delivery (Python, SQL, SAS, Power BI) aligned to accreditation and institutional standards.
- Delivered privacy impact assessments and compliance reviews ensuring responsible, regulation-aligned data use.

Data Analyst & IT Manager
Bridge42 | August 2019 – March 2021 | Perth, Australia

- Delivered automated Power BI dashboards and regression analyses supporting executive and client decision-making.
- Optimised IT infrastructure and vendor strategy, cutting costs 70% while improving scalability and reliability.
- Modernised data-security policies and remote-access infrastructure to NIST-aligned standards during COVID-19.

Education

- Master of Data Science, University of Sydney (studying part-time alongside full-time work).
- Bachelor of Data Science coursework, Curtin University, 2020–2022.

Certifications & Training

- Databricks: AI and LLMOps (2026)
- AWS: Machine Learning Engineering on AWS (Nov 2025)
- LLM-powered Streamlit App with Snowflake (2025)
- Snowflake: MLOps and GenAI integration (2024)